

Karim Elgendi

Washington DC, Virginia



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github.com/kimoiboi

EDUCATION

George Mason University | Fairfax, VA

B.S. in Applied Computer Science, Concentration Software Engineer

May 2023 - December 2026

Relevant Courses: Software Testing/Maintenance, Data Structures, Analysis of Algorithms, Operating Systems, Software Architecture

TECHNICAL SKILLS

Programming Languages: Java, C, SQL/PostgreSQL, HTML, CSS, JavaScript, Dart

Operating Systems: Windows, Linux (Kali, Ubuntu, Pop_OS!)

Technologies/Frameworks & Methodologies: GitHub, Git, Spring/Spring Boot, Agile, Scrum, Flutter, VS Code, Wireshark

PROJECTS

Personal Portfolio Project | Java, JavaScript, Spring Boot, PostgreSQL

January 2026

- Designed and developed a personal portfolio website through project-based learning, applying **HTML**, **CSS**, and **JavaScript** with custom styling and responsive design principles for a polished cross-device experience.
- Built a **Spring Boot REST API** in **Java** with **CRUD endpoints** over a custom **PostgreSQL** database designed from scratch. **Spring Data JPA (Hibernate)** was used for object-relational mapping and persistence.
- Hardened application security with **Spring Security** — hashing admin credentials using **BCrypt**, enforcing brute-force lockout protection against credential-stuffing attacks, and externalizing API keys and secrets through environment variables.
- Deployed and hosted the live site on **Render**, configuring **Cloudflare** as a CDN to cache static assets, reduce origin output bandwidth, and lower latency — gaining hands-on experience with web hosting, caching, and content delivery.

CPU Scheduler Project | C

May 2025

- Developed and implemented core scheduling functions in **C** as a CPU Scheduler that manage process creations based on priority queues, & dealing with termination states within an emulated **Linux** Task Manager.
- Applied bitwise operations, linked list manipulation to streamline processes, & used processing techniques to handle multi-level feedback of queue scheduling & ensuring correct CPU allocation to prevent starvation.
- Gained hands-on experience in memory management, & error-handling practices to achieve functionality with zero memory leaks using **Valgrind**.

EXPERIENCE

Undergraduate Research Assistant | GEORGE MASON UNIVERSITY

February 2026

- Identified and vetted scientific **GitHub** repositories within a defined research domain, assessing each for recency, completeness, and scalability to expand the pool of projects suitable for comparative analysis.
- Performed cross-repository analysis of documentation, onboarding processes, and maintenance practices to surface similarities, differences, and trends in project organization and long-term maintainability.
- Automated much of the data-collection and analysis pipeline by writing **Python** scripts to scrape repository metadata, parse documentation, and aggregate findings.